Mohamad Waleed Nassar

Mechatronics & Automation Engineer

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Exempted My Portfolio MohamadWaleedNassar

PROFILE SUMMARY

Vehicle-electronics engineer at Nissan Egypt owning end-of-line validation (VEP/SIDIS, UDS over CAN) for the PY1B launch: validated 4 variants with Siemens, Nissan Japan and Renault, cut engineering escalations 70%, and built Python tools used daily by ~100 people across Production, Engineering and Aftersales. Wrote a CAN driver in C and built sensor-fusion localization (5 cm accuracy) at Ejad. OEM and Tier-1 experience since 2023. Based in Cairo; open to relocation and international travel.

PROFESSIONAL EXPERIENCE

Nissan Motor Egypt 05/2026 – Present
NPW Specialist Engineer & Vehicle Electronics (VEP/SIDIS) Owner - MSPO
(Manufacturing Strategy Planning Office)
- Selected for a **two-week technical training** assignment at **Nissan Motor UK in May 2026**, focused on its **SIDIS-based VEP system** and production practices, Utilized its outputs in NMEG, by implementing new Systems & Optimizations.
- Cut engineering escalations for vehicle-electronics concerns by **70%** by building a Python/API dashboard on the SIDIS database that shows live VEP status per vehicle, DTC alerts and historical analysis for production and engineering.
- Eliminated **10+** sequence/variant mismatches between the production plan and actual build by developing PVEDsync, a Python/API link between VEP and Production Control.
- Gave Aftersales dealers ECU configuration, software and programming records for **3000+** vehicles by building an API-based traceability portal on manufacturing data.
- Closed **9** NPW gaps across **4** plant shops, moving the audit score from **80%** to **95%**, by leading 4 foremen through gap analysis and Kaizen plans.

Nissan Motor Egypt 07/2025 – 05/2026
PE Assembly Engineer - (VEP ownership continued under MSPO)
- Launched **VEP** (Vehicle Electronic Process) end-of-line validation for the **PY1B new model**, replacing legacy validation across Assembly, by deploying hardware and software and validating 4 variants (function checks, MIC screen logic, DTC behavior).
- Resolved **200+** recurring end-of-line failures across **14** ECUs and 4 PY1B variants by root-cause analysis of UDS-over-CAN diagnostic traffic (DTC read/clear, DID read/write, routine control), cutting EOL re-tests by 30%.
- Coordinated with **Siemens** and **global Nissan** and Renault engineering teams to resolve **6 main** system integration issues and finalize operating parameters.
- Secured Trim & Chassis launch readiness by building the PY1B Marshaling List covering **1000+** operations at **24** stations (V5+ revisions through PT builds), cutting part-shortage or line-stop events by **60%**.

Ejad 10/2024 – 06/2025
Graduation Project "Embedded Systems BLE Localization in Vehicles"
- Enabled real-time CAN communication between NXP anchors and Tiva-C at **5 ms** latency and **2%** message loss by writing the CAN driver in C.
- Improved localization accuracy to 5 cm by implementing particle-filter **sensor fusion** of RSSI data from multiple BLE anchors.
- Contributed to OSEK-based task management and ESP32 server connectivity module.

Nissan Motor Egypt 07/2024 – 12/2024
Production Engineering Intern
- Cut manual interventions by **40%** by integrating Atlas Copco tightening tools with PLCs and control circuits (QART) in the Assembly shop.

LEONI EGYPT 08/2023 – 09/2023
Production Concept Department Intern
- Contributed to a floor-area reduction across 10 **LEONI factories** by analyzing and creating optimization opportunities.
- Gained hands-on exposure to industrial operations, lean manufacturing concepts, and plant layout planning.

EDUCATION

Mechatronics & Automation Engineering Bachelor degree 2020 – 06/2025 | Cairo, Egypt
(GP: Excellent, GPA: Very Good)
Ain Shams University & University of East London
- Focused on embedded systems, automation, robotics, and control engineering. Completed coursework and other projects in microcontrollers, PLC programming, and full mechatronic system design.

Data Science Diploma (14 month) 05/2023 – 09/2024
Alx
- Trained in Python programming, data wrangling, statistical analysis, machine learning algorithms, data visualization, and applied predictive modeling using real-world datasets.

Embedded Systems Diploma 06/2023 – 12/2023
AMIT
- Learned embedded systems fundamentals and microcontroller architecture; developed complete drivers, memory-mapped interfaces, and hands-on projects using AVR microcontrollers to integrate and control various modules.

EXTRA-CURRICULAR ACTIVITIES

AIESEC in Ain Shams University 02/2024 – 02/2026
Vice President and Head of Finance & Legalities

IEEE ASUSB 06/2023 – 12/2024
Head of Multimedia

ASU Racing Team 09/2021 – 10/2022
Head of Multimedia Department

ISLC - AUC (American University in Egypt) 08/2019 – 03/2020
Juniors Team

Tedx Manarat Al-Farouk School 09/2017 – 04/2019
Fund Raiser & Marketer

COURSES

McKinsey Forward Program 04/2025 – 2026
McKinsey Academy
Developing must-have skills to navigate my career journey in the future of work successfully through the digital, self-paced program.

AUTOSAR basics 07/2024 – 08/2025
Garraio
Learned the fundamentals of the AUTOSAR architecture, its layered structure, and its role in standardizing automotive software development. Covered key concepts such as application and service layers, ECU abstraction, and real-world use cases in embedded automotive systems.

Computer Vision & Machine Learning 10/2024 – 11/2024
UDEMY
Completed a course covering core concepts in computer vision and machine learning, including image processing, feature extraction, and supervised learning algorithms. Applied practical techniques for data analysis and pattern recognition.

Python Programming Bootcamp 09/2024 – 10/2024
UDEMY
Completed an intensive course covering Python fundamentals, data structures, libraries, and practical programming skills. Gained experience writing clean, efficient code and using Python for automation and problem-solving.

ROS for Robotics Development 07/2023 – 08/2023
Ain Shams University
Gained hands-on experience with Robot Operating System (ROS), covering core concepts like nodes, topics, and launch files. Learned to simulate robotic systems in Gazebo, import custom designs, and implement motion planning and control using MoveIt for robotic arms.

Basic Automation Systems using PLC Training 07/2022 – 08/2022
SCHNIDER ELECTRIC
- Gained hands-on experience with PLC types and ladder programming through real industrial applications.
- Designed and simulated a complete factory line to test and validate custom PLC logic in a production-like environment.
- Built a solid foundation in industrial automation and control systems.

SOFTWARE & DATA INTEGRATION

Power BI • MATLAB • Proteus Simulation • Solidworks • Autodesk Inventor • AVR Studio / Microchip Studio • SAM • AutoCAD • ANSYS • TIA Portal • 3D-MAX • Adobe Photoshop • Microsoft Office

PROGRAMMING LANGUAGES

C/C++ | Python | Java | HTML&CSS

SKILLS

- Vehicle Electronics & Manufacturing
- Leadership
- Analytical and Problem-Solving skills
- Continuous Improvement (Kaizen)
- SIDIS
- SIDIS & End-of-Line Validation
- DTC Analysis
- New Model Launch Support
- UDS (ISO 14229) & CAN

LANGUAGES

- Arabic
- English
- French

INTERESTS

Photography | Doodle Arts | Innovation & Technology | Poetry | Travelling & Exploring | Hiking & Camping

MAIN PROJECTS

Vehicle Electronic Process (VEP) System Deployment at Nissan Egypt 08/2025 – Present

"New model vehicle electronics deployment, end-of-line validation, and continuous improvement"

Led the complete deployment of the **VEP (Vehicle Electronic Process)** system for the PY1B model, replacing Old validation methods and standardizing end-of-line quality checks across the Assembly shop. The project included hardware installation, software configuration, SOS creation, and execution of a 4-vehicle variant validation trial.

- Developed **SOSs, work instructions, and training materials** for operators, team leaders, and engineering staff.
- Conducted end-to-end **system validation**, including function checks, MIC screen logic verification, and DTC behavior confirmation.
- Coordinated with **Siemens, NML Japan and RNTBCI** (India) & Renault to resolve integration issues and finalize system parameters.

This project established **VEP as the new Nissan Egypt standard,** improving traceability, validation reliability, and process discipline across Assembly.

VEP Mega Dashboard 05/2026 – Present

"Vehicle Electronics Monitoring & Diagnostics"

Designed and implemented a centralized dashboard integrating with the SIDIS database through Python backend services and APIs. The platform brings together vehicle production monitoring, diagnostic knowledge, alerts, and historical analysis in an intuitive interface for shop-floor and engineering users.

- Translated SIDIS data into an accessible frontend designed for daily use by production and engineering teams.
- Built an Arabic-English troubleshooting dictionary documenting previously encountered issues, DTCs, and corrective guidance, enabling shop-floor teams to resolve recurring concerns with less reliance on engineering support.
- Developed real-time monitoring of produced vehicles and their VEP processing status to support production visibility.
- Implemented alerting and analytical modules to highlight issues requiring attention and support follow-up.
- Centralizes vehicle electronics information and practical troubleshooting knowledge to support consistent production.

PVEDsync 08/2026 – Present

"Vehicle sequence, and processing-status visibility for Production Control"

Developed an integration between the VEP system and Production Control to track vehicles, their variants, and processing states. The solution retrieves data from the vehicle electronics data server using Python scripts and APIs and presents it through a friendly UI.

- Enabled teams to check vehicle progression against the intended production sequence and variant requirements.
- Provided visibility into sequence and status discrepancies to support investigation and resolution of production misalignment.

Nissan Aftersales Vehicle Electronics Traceability Portal 09/2026 – Present

"Controlled access to ECU config, SW, and vehicle programming records"

Developed an API-based portal providing authorized Aftersales personnel with vehicle electronics and ECU traceability information captured during manufacturing. The solution presents technical records through an intuitive interface to support dealer-side troubleshooting and vehicle repair.

Marshaling List Development & New Model Operation Distribution for PY1B 08/2025 – 02/2026

"Full documentation of Trim & Chassis operations and station-level part allocation for new model launch"

Developed the complete **Marshaling List** for the new model, covering all Trim & Chassis shop operations. The project ensured accurate part allocation, correct quantities, and balanced distribution of operations across stations to secure production readiness.

- Documented **every new model operation** occurring along the Trim & Chassis line, validating technical feasibility for each station.
- Assigned and verified **all part numbers, quantities, tool requirements, and installation sequences** for each operation.
- Collaborated with SCM, Production, and PE teams to confirm material availability, installation feasibility, and ergonomics.
- Updated multiple Marshaling List versions (V5+) during PT builds, reflecting design changes and process adjustments.

This project became the foundation for correct new model execution in Assembly and ensured alignment between Engineering, Production, and SCM during the launch phase.

Phone as a Key Using BLE in EJAD 10/2024 – 06/2025

"Advanced Vehicle Access & Localization System"

Contributed in the design and implementation of a secure vehicle access system leveraging Bluetooth Low Energy (BLE) 5.3 technology to replace traditional car keys with smartphones. Developed a system integrating multiple BLE anchors around the vehicle for high-accuracy localization using Several technologies like **RSSI**, **Channel Sounding**. Designed and implemented a fusion processor for real-time localization data processing, device authentication, and zone-based vehicle control. This solution advances next-generation vehicle access, providing high accuracy, scalability, and compatibility with existing BLE infrastructure.

Production Line with Prismatic Robotic Arm 02/2023 – 06/2023

"Automated Sorting System with Vacuum and Pneumatic Mechanisms"

In this project, completed as part of the Design of Mechatronics 1 course, a team of three designed a fully automated production line featuring a prismatic robotic arm, a sucking mechanism, and a feeding mechanism. The process began with products being manually placed in a magazine. The feeding mechanism, based on a pneumatic cylinder, pushed the products from the magazine to a classification station that identified whether the product was made of metal or plastic. The robotic arm, operating along three axes, then utilized a vacuum generator to pick up the product and place it in the appropriate area based on its material.

The project leveraged FreeRTOS for task management, involved the use of multiple Arduinos, and included the design and fabrication of a complete PCB board. The project was fully simulated in MATLAB, with the mechanical design completed using Autodesk Inventor. This project marked the team's first significant foray into large-scale mechatronics systems.